PV101 DAY 5

Commissioning, Protection & Fault Finding

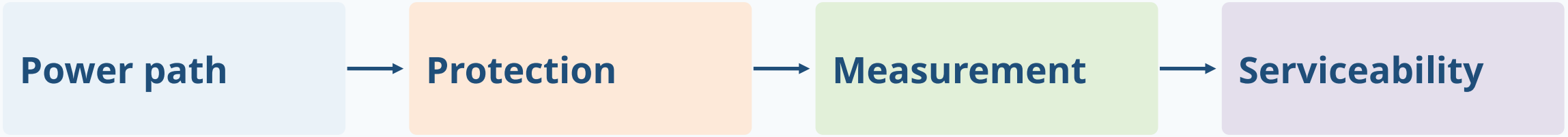
Monday, July 13, 2026 | CTS Cohort 6



Verify the path before you energize it.

Today’s job

Use yesterday’s completed build as a system to inspect, prove, protect, and troubleshoot.



- Students must identify where current can come from.
- Fuses protect wires from too much current.
- Every important claim gets verified with a meter.
- The final build should make sense to a tired future technician.

The four numbers

A small system is easier when students can translate the electrical language.

Voltage	Current	Power	Resistance
V	**A**	**W**	**Ω**
Electrical pressure	Flow of charge	Work being done	Opposition to flow

Power formula: $P = V \times A$

Example: 12.8 V × 5 A = 64 W

Series vs. parallel

Ask students to predict first, then prove it.

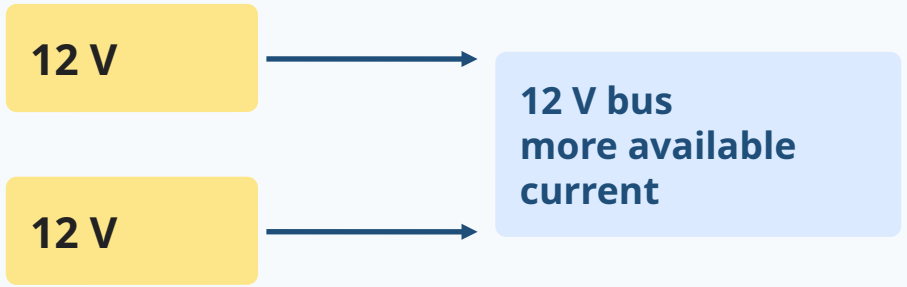
Series: voltage adds



Classroom sentence: series changes voltage, parallel keeps voltage common.

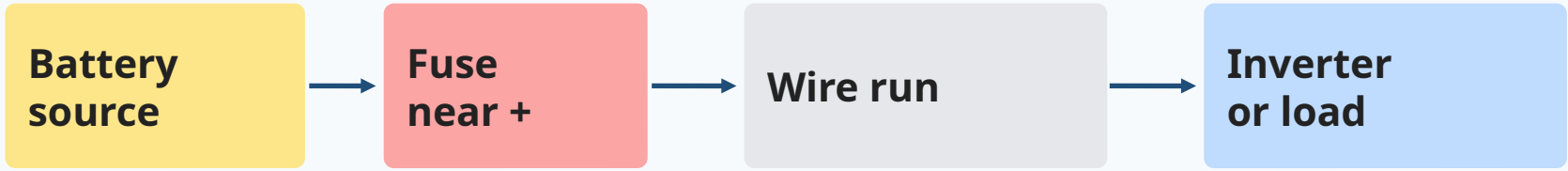
- Same path for current
- One weak part affects the whole string

Parallel: voltage stays, current capacity increases



Fuse logic

A fuse protects the conductor from the source. It is not a magic device shield.

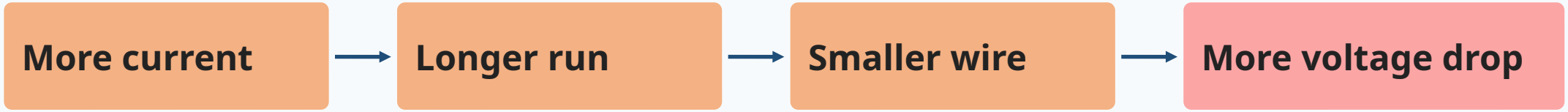


If the wire shorts before the fuse, the fuse cannot help. Humanity learns this repeatedly.

- Place the fuse close to battery positive.
- Fuse size must match the wire and expected load.
- Stop and inspect if heat, smell, sparks, or weird readings appear.

Wire sizing and voltage drop

The wire is part of the system, not just a thing that connects the system.



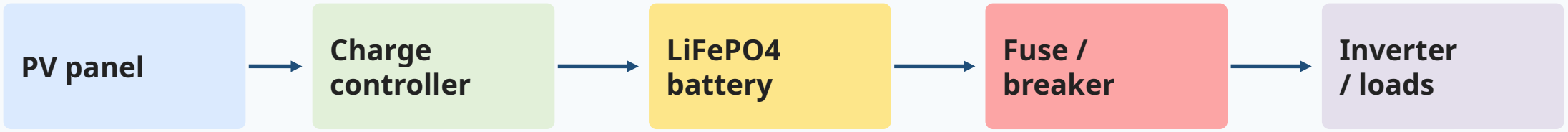
Voltage drop test

- Measure voltage at battery under load.
- Measure voltage at load/inverter under load.
- Subtract: battery-end minus load-end.
- High drop means resistance, undersized wire, bad crimp, loose terminal, or too much current.

Never “fix” voltage drop by just turning things up. Find the cause.

RV solar power path

Students should be able to draw this before touching wires.



Inspection questions

- Where can current come from?
- What protects each wire?
- What readings prove polarity and voltage?
- What would you label for service later?

Order matters

- Controller to battery first unless manual says otherwise.
- PV last. Loads after inspection.
- Unexpected readings mean stop and trace.

Afternoon lab: commission, then diagnose

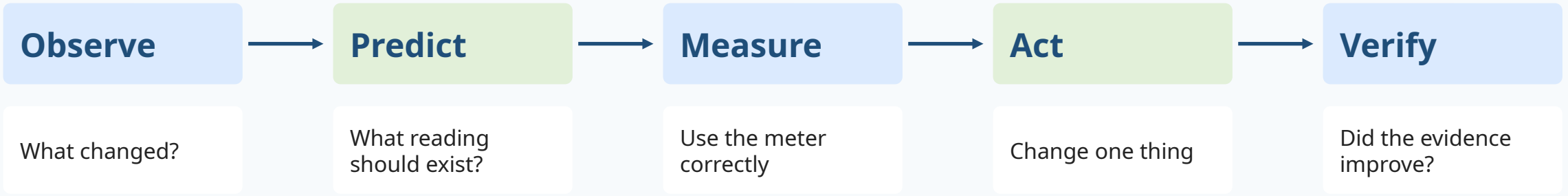
Keep Day 4 systems assembled. Rotate through inspection, measurement, fault evidence, and restoration.

Lead inspector	Meter lead	Recorder / verifier
Traces path	Predicts readings	Records evidence
Checks protection	Measures safely	Controls sequence
Improves labels	Checks voltage drop	Verifies restoration

No rebuilding unless a measured fault requires a correction. One change at a time.

Commissioning and troubleshooting flow

No random touching. Observe, predict, measure, act. Wild concept, apparently.



- Wrong polarity? Stop before energizing.
- No output? Check source, fuse, wiring path, and controller state.
- Low voltage under load? Check current, wire size, crimps, and voltage drop.
- Heat, smell, spark, or weird reading? Hands off, call instructor.

Exit ticket

Three things every student should leave with.

I can verify with a meter…

A fuse protects…

A fault I can now diagnose systematically…

Cleanup standard

Every kit restored and verified. Tools back. No mystery wires. Labels improved for the next technician.

Instructor close

The win is not rebuilding yesterday’s work. The win is proving it, diagnosing it, and returning it to known-good condition.